NINAD NAIK

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EDUCATION

Bachelor of Engineering in Information Technology 2021 - 2025
University of Mumbai, Thadomal Shahani Engineering College (TSEC), Mumbai, India CGPA: **8.54 / 10**

WORK EXPERIENCE

Associate Backend Developer, *Commotion* (acq. by Tata Communications) (gocommotion.com) Jan 2025 - Sep 2026

- Promoted from Backend Developer Intern (January 2025 - June 2025)
- Engineered **low-latency backend architectures** using NestJS and FastAPI, utilizing targeted database indexes and query optimization to handle **high-concurrency enterprise workloads**.
- Developed **MCP and A2A orchestration layer** for real-time contextual voice and text AI agents.
- Built robust **AI integrations** with platforms like Salesforce, ServiceNow, and SAP, and 100+ others, unifying workflows.
- Enforced security using OAuth2.0 and JWT auth schemes, ensuring platform integrity via SAST security analysis.
- Developed **low-latency Voice AI** solutions in collaboration with **multiple Tata companies and MNCs** to streamline automated operations and enhance customer experience.

Full Stack Developer Intern, *Arxena Inc.* (arxena.com) June 2024 - July 2024

- Built a **GPT-powered WhatsApp recruitment chatbot** in **NestJS, GraphQL and WebSocket API** that screened candidates on JD-based questions, shared JD PDFs, stored resumes in **Amazon S3** and transcribed voice answers, with a WhatsApp chat UI in **React.js** inside **Twenty CRM** showing each candidate's CRM status.
- Reduced **AWS EC2** costs by up to **75%** by moving builds off the live server to on-demand instances from an **AMI** (details).

OPEN SOURCE MENTORSHIP

The Linux Foundation - LFX - Linux Kernel Mentorship Program mentee, Spring 2026 (link) March 2026 - August 2026

- Graduated with **11 patches** contributed to Linux Kernel mainline; published a blog sharing my experience.
- Converted text-based **Devicetree bindings** to JSON Schema; fixed documentation links, typos, and driver error handling.
- Worked with **kernel development workflow**, **patch submission**, and **review cycles** and used tools like **cscope**, **Coccinelle**, and **smatch** for static analysis. Selected for the mentorship program from **693 applicants**.

PROJECTS

Redspot, *FastAPI, React.js, Clickhouse, Kafka, PostgreSQL, Docker* | (GitHub) June 2026

- Built a **real-time website click analytics platform** with a FastAPI ingestion server, Kafka message queue, and ClickHouse storage pipeline that captures user interactions via a lightweight JavaScript tracking script.
- Developed a React dashboard that renders target websites in an iframe and overlays a **color-coded heatmap visualization of click density** using canvas, powered by live data queried from ClickHouse

SpeakSure, *Python, Flask, Tensorflow, React.js, MongoDB* | (Live app, GitHub) October 2025

- Developed an **Automated AI Behavioural Interview system** that quantifies **confidence level** in candidate's speech and provides AI insights to the hiring manager. Presented **research paper** at ICCIS 25 based on new models and findings.
- Trained **multilayer perceptron (MLP)** and **Fuzzy Hypersphere Neural Network** models on a custom made dataset with **751** audio samples of 30 seconds each by extracting features with Wav2Vec2 and achieved **96%** relaxed testing accuracy.

Shortomega, *React.js, Next.js, Node.js, NestJS, TypeScript, Redis, Docker* | (Live app, GitHub) November 2024

- Developed a **Full Stack URL Shortening App** that generates short URLs for long URLs, with a focus on system design.
- Published a technical blog describing the methods of using **Redis as a Primary Database**.
- Implemented **Dashboard, Analytics data aggregation and Rate Limiting** features, and **Dockerized** the application.

SKILLS

Languages: TypeScript, JavaScript, Python, Go, C, C++, Java
Backend & APIs: NestJS, FastAPI, Node.js, GraphQL, gRPC
AI: MCP, A2A, RAG, LangChain, LangGraph, Agentic AI
Web: React.js, Tailwind CSS, HTML, CSS, Next.js, HeroUI
Data & Messaging: PostgreSQL, MongoDB, Redis, Kafka
Platforms & Tools: Linux, Docker, Kubernetes, AWS

RESEARCH AND PUBLICATION

Evaluating Human Speech Confidence with Deep Learning Techniques, **Ninad Naik, et al.** September 2025

- Validated **Multi Layer Perceptron**, **Fuzzy Hypersphere Neural Network** and **Logistic Regression** classifiers over **MFCC** and **Wav2Vec2** features on a custom **751-recording** dataset using **5-fold stratified cross-validation**, reaching **96% relaxed accuracy** across 5 confidence levels.
- Presented at **ICCIS 2025**, BITS Pilani, Goa; accepted for publication in SCOPUS-indexed **Springer** LNNS. (Certificate)

ADDITIONAL INFORMATION

- Open Source Contribution to the Linux Kernel, n8n.io and CircuitVerse.
- First Runner Up (CSS Battle 2022 & Javagenix 2022), 2nd Runner Up (PyExpo 2023), Top 10 (ACE Hacks 2024)
- Maintaining a **tech blog** to share experiences, learnings and results on tech.ninadnaik.me